Lab 5: Data Visualization and Interpretation using Seaborn

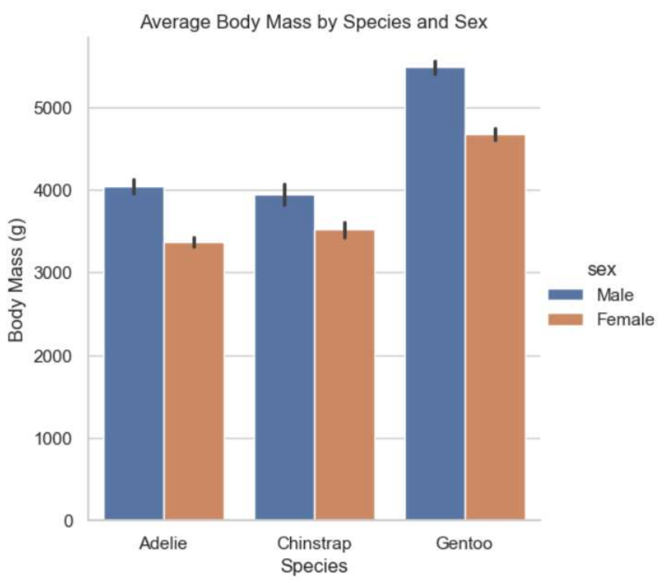
Imports

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
sns.set(style="whitegrid")
```

Dataset 1: Penguins Dataset

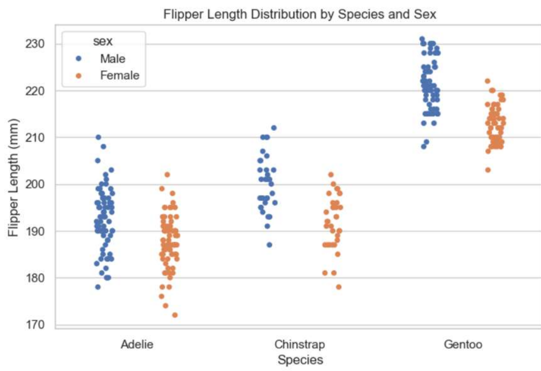
Task 1: Catplot – Average Body Mass by Species and Sex

```
penguins = sns.load_dataset("penguins")
sns.catplot(data=penguins, kind="bar", x="species", y="body_mass_g", hue="sex")
plt.title("Average Body Mass by Species and Sex")
plt.xlabel("Species")
plt.ylabel("Body Mass (g)")
plt.show()
```



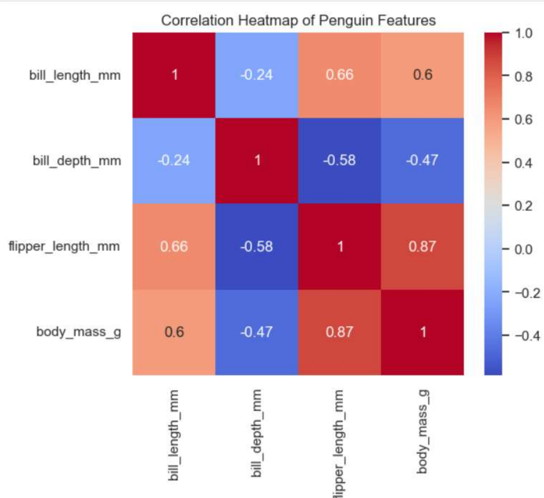
Task 2: Strip Plot – Flipper Length Distribution

```
plt.figure(figsize=(8,5))
sns.stripplot(data=penguins, x="species", y="flipper_length_mm", hue="sex",
dodge=True)
plt.title("Flipper Length Distribution by Species and Sex")
plt.xlabel("Species")
plt.ylabel("Flipper Length (mm)")
plt.show()
```



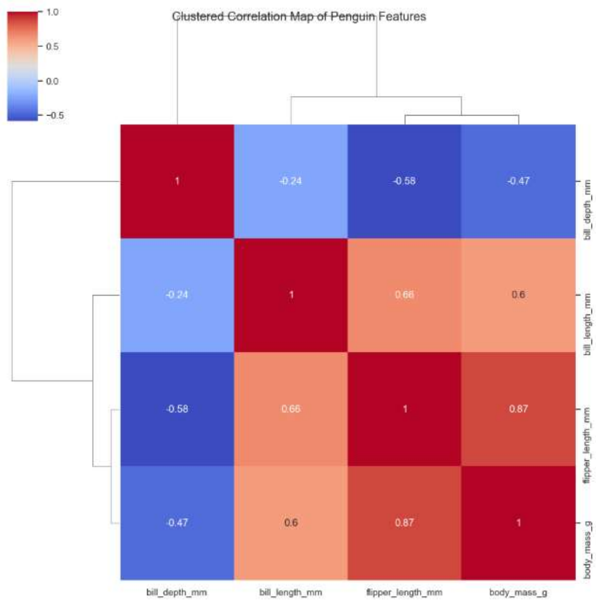
Task 3: Heatmap – Correlation Matrix

```
corr =
penguins[['bill_length_mm','bill_depth_mm','flipper_length_mm','body_mass_g']].
corr()
plt.figure(figsize=(6,5))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap of Penguin Features")
plt.show()
```



Task 4: Clustermap – Clustered Correlation Map

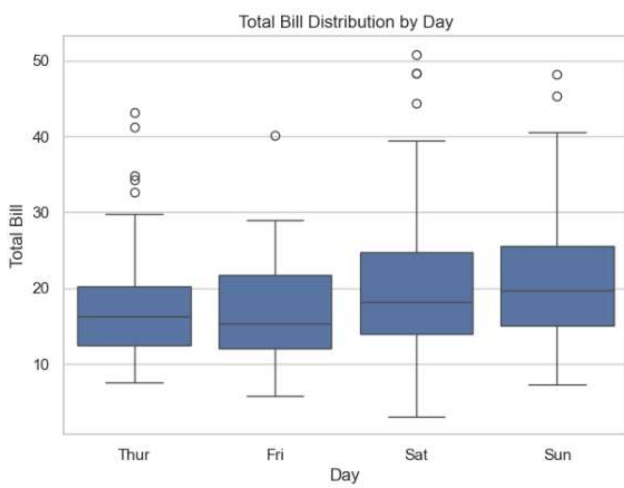
```
sns.clustermap(corr, annot=True, cmap="coolwarm")
plt.suptitle("Clustered Correlation Map of Penguin Features")
plt.show()
```



Dataset 2: Tips Dataset

Task 5: Box Plot – Total Bill Distribution by Day

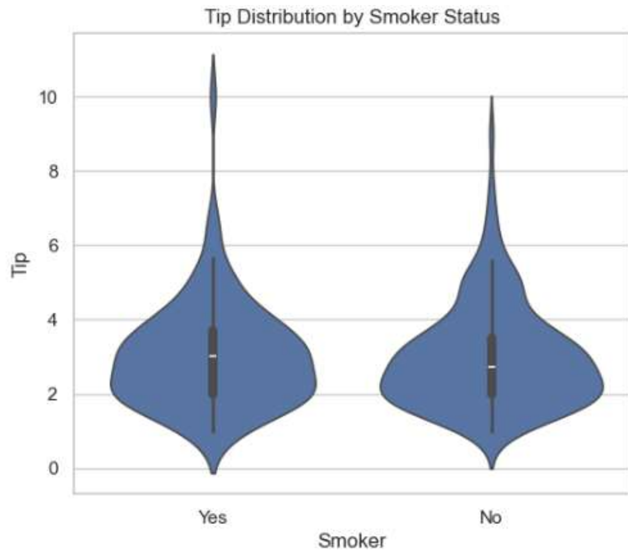
```
tips = sns.load_dataset("tips")
plt.figure(figsize=(7,5))
sns.boxplot(data=tips, x="day", y="total_bill")
plt.title("Total Bill Distribution by Day")
plt.xlabel("Day")
plt.ylabel("Total Bill")
plt.show()
```



Task 6: Violin Plot – Tip Distribution by Smoker Status

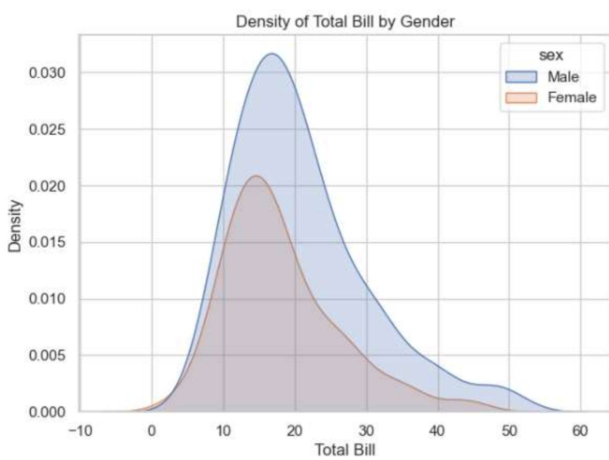
```
plt.figure(figsize=(6,5))
sns.violinplot(data=tips, x="smoker", y="tip")
plt.title("Tip Distribution by Smoker Status")
plt.xlabel("Smoker")
```

```
plt.ylabel("Tip")
plt.show()
```



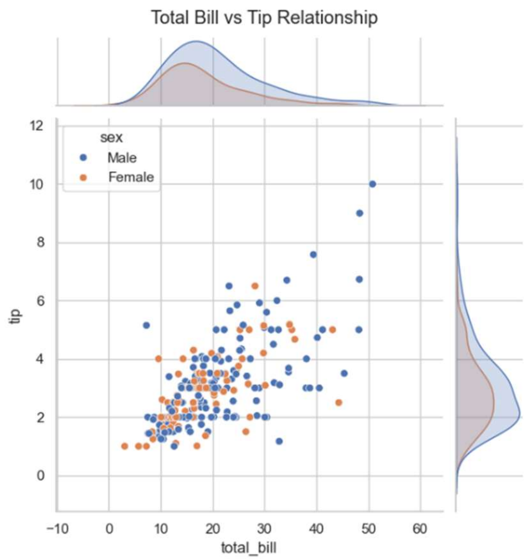
Task 7: KDE Plot – Density of Total Bill by Gender

```
plt.figure(figsize=(7,5))
sns.kdeplot(data=tips, x="total_bill", hue="sex", fill=True)
plt.title("Density of Total Bill by Gender")
plt.xlabel("Total Bill")
plt.show()
```



Task 8: Joint Plot – Total Bill vs Tip

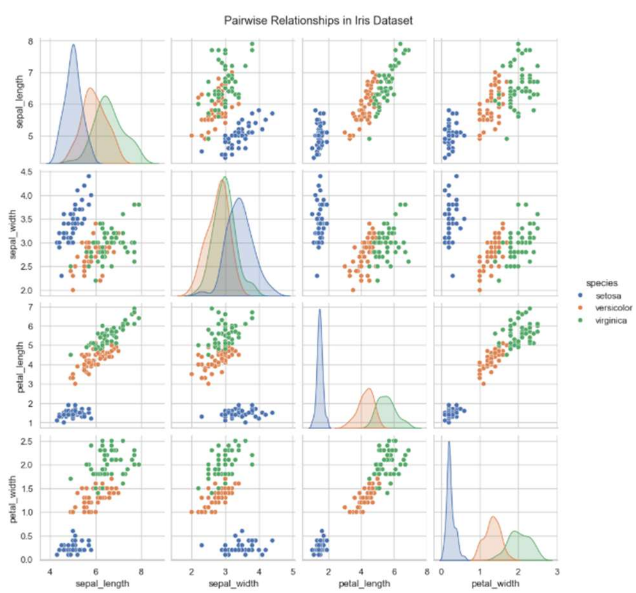
```
sns.jointplot(data=tips, x="total_bill", y="tip", hue="sex")
plt.suptitle("Total Bill vs Tip Relationship", y=1.02)
plt.show()
```



Dataset 3: Iris Dataset

Task 9: Pairplot – Pairwise Relationships

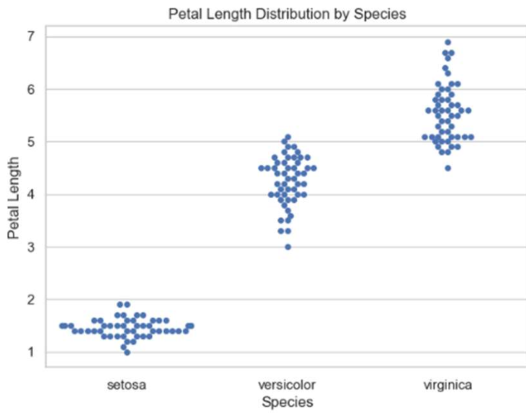
```
iris = sns.load_dataset("iris")
sns.pairplot(iris, hue="species")
plt.suptitle("Pairwise Relationships in Iris Dataset", y=1.02)
plt.show()
```



Task 10: Swarm Plot – Petal Length Distribution

```
plt.figure(figsize=(7,5))
sns.swarmplot(data=iris, x="species", y="petal_length")
plt.title("Petal Length Distribution by Species")
plt.xlabel("Species")
```

```
plt.ylabel("Petal Length")
plt.show()
```



Task 11: Bubble Chart – Sepal vs Petal Dimensions

```
plt.figure(figsize=(7,5))
sns.scatterplot(data=iris, x="sepal_length", y="sepal_width",
                size="petal_length", hue="species", sizes=(20,200))
plt.title("Sepal vs Petal Dimensions in Iris")
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.show()
```